Advanced Stochastic Calculus with Applications

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1 Preliminaries about Stochastic Processes

Intermission 1

The purpose of this section is to introduce the concepts of **adapted** and **progressively measurable** processes, discuss their relationship, and present key results related to stopping times.

These are foundational for our later study of **stochastic control** and **optimal stopping**.

Adaptedness is crucial because it ensurs our models respect causality. For instance, in an American option exercise strategy, a decision at time t cannot be contingent on a future price $X_s, s > t$, as that information is unknown.

Progressive Measurability is a more technical requirement that arises in continuous time. It is a prerequisite for stochastic integration, ensure the process is "well-behaved" over time intervals.

You may view it as "path-adaptedness": while adaptedness ensures we know the process' value at a specific point, progressive measurability ensures that the entire history of the process up to t is mathematically tractable as a single object.

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be the probability space on which we're going to work.

Let $\mathbb{X} = (X_t)_{t\geq 0}$ be a stochastic process which is *measurable*, *adapted*, and *progressively measurable*.

Definition 2

A stochastic process is $(t, \omega) \to X_t(\omega)$ is $\mathcal{B}(\mathbb{R}_+) \times \mathcal{F}$**-measurable** if

$$\{(t, \omega) : X_t(\omega) \in A\} \in \mathcal{B}(\mathbb{R}_+) \times \mathcal{F}, \forall A \in \mathcal{B}(\mathbb{R}^d)$$

Definition 3

A stochastic process $\mathbb{X}$ is **adapted to** $(\mathcal{F}_t)_{t\geq 0}$ if $X_t \in m\mathcal{F}_t, \forall t \geq 0$.

Note that $m\mathcal{F}_t$ denotes the set of measurable functions with respect to the σ-algebra $\mathcal{F}_t$ in the filtration $(\mathcal{F}_t)_{t\geq 0}$.

Definition 4

A stochastic process $\mathbb{X}$ is **progressively measurable** if $\forall t \geq 0, \{(s, \omega) \in [0, t] \times \Omega : X_s(\omega) \in A\} \in \mathcal{B}([0, t]) \times \mathcal{F}_t, \forall A \in \mathcal{B}(\mathbb{R}^d)$.

Homework 5

Show that if $\mathbb{X}$ is progressively measurable, then it is also adapted.

Hint: use the properties of measurable functions for the product σ-algebra.

Proposition 6

If $\mathbb{X}$ is adapted and right-continuous, then it is progressively measurable.

Proof

Fix $t > 0$, and define:

$$X_s^n = \begin{cases} X_{\frac{k+1}{2^n}t}, & s \in \left[\frac{k}{2^n}t, \frac{k+1}{2^n}t\right) \\ X_t, & s = t \end{cases}$$

> **Homework 7**
> Show that $X_s(\omega) = \lim_{n\to\infty} X_s^n(\omega), \forall (s,\omega) \in [0,t] \times \Omega$.

If we show that $(s,\omega) \to X_s^n(\omega)$ is $\mathcal{B}([0,t]) \times \mathcal{F}_t$-measurable then $(s,\omega) \to X_s(\omega)$ is also $\mathcal{B}([0,t]) \times \mathcal{F}_t$-measurable by pointwise limit.

Since $t > 0$ is arbitrary, we conclude that $\mathbb{X}$ is progressively measurable.

Fix $A \in \mathcal{B}(\mathbb{R}^d)$: $\{(s,\omega) \in [0,t] \times \Omega : X_s^n(\omega) \in A\}$

$$= \bigcup_{k=0}^{2^n-1} \left(\left[\frac{k}{2^n}t, \frac{k+1}{2^n}t\right) \times \left\{\omega \in \Omega : X_{\frac{k+1}{2^n}t}(\omega) \in A\right\}\right) \bigcup (\{t\} \times \{\omega \in \Omega : X_t(\omega) \in A\})$$

Then:

$$\{(s,\omega) \in [0,t] \times \Omega : X_s^n(\omega) \in A\} \in \mathcal{B}([0,t]) \times \mathcal{F}_t$$

■

Remark 8. *If $\mathbb{X}$ is measurable and $\tau : \Omega \to [0,+\infty)$ is $\mathcal{F}$-measurable, then $\omega \to X_{\tau(\omega)}$ is $\mathcal{F}$-measurable as composition of measurable maps.*

Adopt the notation $X_t(\omega) = X(t,\omega)$ so that $X^{-1}(A) = \{(t,\omega) : X(t,\omega) \in A\}$. Then, $\{\omega \in \Omega : X(\tau(\omega),\omega) \in A\} = \left\{\omega \in \Omega : (\tau(\omega),\omega) \in \underbrace{X^{-1}(A)}_{\in \mathcal{B}(\mathbb{R}_+)\times\mathcal{F}}\right\}$.

Notice that $\omega \to (\tau(\omega),\omega)$ is a mapping $\Omega \to \mathbb{R}_+ \times \Omega$. For any $E \in \mathcal{B}(\mathbb{R}_+)$ and $F \in \mathcal{F}$ we have $\{\omega \in \Omega : (\tau(\omega),\omega) \in E \times F\} = \{\omega \in \Omega : \tau(\omega) \in E\} \cap F \in \mathcal{F}$.

This extends to $\{\omega : (\tau(\omega),\omega) \in B\} \in \mathcal{F}, \forall B \in \mathcal{B}(\mathbb{R}_+) \times \mathcal{F}$ because $\{E \times F, E \in \mathcal{B}(\mathbb{R}_+), F \in \mathcal{F}\}$ generates $\mathcal{B}(\mathbb{R}_+) \times \mathcal{F}$.

In the definition below, let Π be either $[0,+\infty)$ or $\mathbb{Z}^+$.

> **Definition 9**
> Given a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\Pi}, \mathbb{P})$ and a stopping time τ, we define
>
> $$\mathcal{F}_\tau := \{A \in \mathcal{F}_\infty : A \cap \{\tau \le t\} \in \mathcal{F}_t, \forall t \in \Pi\}$$

> **Homework 10**
> Show that $\mathcal{F}_\tau$ is a σ-algebra on Ω.

Proposition 11

Let σ, τ be stopping times. Then:

1. $\tau \in m\mathcal{F}_\tau$
2. $\tau \wedge \sigma$ and $\tau \vee \sigma$ are stopping times
3. $\sigma(\omega) \leq \tau(\omega), \forall \omega \in \Omega \implies \mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$
4. $\mathcal{F}_{\sigma \wedge \tau} = \mathcal{F}_\sigma \cap \mathcal{F}_\tau$

Proof

(ii) was proven in PFF and it's left as HW.

(i) It suffices to show $\{\omega : \tau(\omega) \leq s\} \in \mathcal{F}_\tau, \forall s \in \Pi$. By def $\{\tau \leq s\} \in \mathcal{F}_s \subseteq \mathcal{F}_\infty$. Fix $t \in \Pi$. Then

$$\{\tau \leq s\} \cap \{\tau \leq t\} = \{\tau \leq s \wedge t\} \in \mathcal{F}_{s \wedge t} \subseteq \mathcal{F}_t$$

(iii) Take $F \in \mathcal{F}_\sigma$. Then $F \in \mathcal{F}_\infty$ and $F \cap \{\sigma \leq t\} \in \mathcal{F}_t, \forall t \in \Pi$.

$$F \cap \{\tau \leq t\} = F \cap \underbrace{\{\sigma \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{\tau \leq t\}}_{\in \mathcal{F}_t} \quad \text{because } \sigma \leq \tau$$

Then $F \in \mathcal{F}_\tau$.

(iv) Take $F \in \mathcal{F}_{\sigma \wedge \tau}$. Then $F \in \mathcal{F}_\infty$ and $F \cap \{\sigma \wedge \tau \leq t\} \in \mathcal{F}_t, \forall t \in \Pi$. Now, $\forall t \in \Pi$,

$$F \cap \{\tau \leq t\} = F \cap \underbrace{\{\sigma \wedge \tau \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{\tau \leq t\}}_{\in \mathcal{F}_t}$$

$\implies F \in \mathcal{F}_\tau$ and by the same argument $F \in \mathcal{F}_\sigma$. Thus, $\mathcal{F}_{\sigma \wedge \tau} \subseteq \mathcal{F}_\tau \cap \mathcal{F}_\sigma$.

Take $F \in \mathcal{F}_\tau \cap \mathcal{F}_\sigma$. Then $F \in \mathcal{F}_\infty$, $F \cap \{\tau \leq t\} \in \mathcal{F}_t$ and $F \cap \{\sigma \leq t\} \in \mathcal{F}_t, \forall t \in \Pi$. Now, for $t \in \Pi$,

$$\begin{aligned} F \cap \{\tau \wedge \sigma \leq t\} &= (F \cap \underbrace{\{\tau \leq t\}}_{\in \mathcal{F}_t} \cap \{\tau \wedge \sigma \leq t\}) \cup (F \cap \{\tau \wedge \sigma \leq t\} \cap \{\tau > t\}) \\ &= \underbrace{F \cap \{\sigma \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{\tau > t\}}_{\in \mathcal{F}_t} \end{aligned}$$

$\implies F \in \mathcal{F}_{\tau \wedge \sigma} \implies \mathcal{F}_\tau \cap \mathcal{F}_\sigma \subseteq \mathcal{F}_{\tau \wedge \sigma}$. ■

Proposition 12

If τ is a stopping time and $\mathbb{X}$ is progr. meas. then $X_\tau \in m\mathcal{F}_\tau$.

Proof

We must prove that $\forall A \in \mathcal{B}(\mathbb{R}^d)$, $\{\omega \in \Omega : X_\tau(\omega) \in A\} \in \mathcal{F}_\tau$. Clearly $\{\omega \in \Omega : X_\tau(\omega) \in A\} \in \mathcal{F}_\infty$. Now, for any $t \geq 0$,

$$\{X_\tau \in A\} \cap \{\tau \leq t\} = \{X_{\tau \wedge t} \in A\} \cap \{\tau \leq t\}$$

The random variable $\tau \wedge t : \Omega \to [0, t]$ is $\mathcal{F}_t$-measurable because, $\forall s \geq 0$,

$$\{\tau \wedge t \leq s\} = \begin{cases} \Omega & \text{if } s \geq t \\ \{\tau \leq s\} & \text{if } s \in [0, t] \end{cases} \in \mathcal{F}_t$$

Now, let $X^t(s, \omega) := X_{s\wedge t}(\omega)$. Then $\forall A \in \mathcal{B}(\mathbb{R}^d)$, $\{\omega \in \Omega : X_{\tau\wedge t}(\omega) \in A\} = \{\omega \in \Omega : X^t(\tau(\omega), \omega) \in A\}$

$$= \left\{ \omega \in \Omega : (\tau(\omega), \omega) \in \underbrace{(X^t)^{-1}(A)}_{\in \mathcal{B}([0,t])\times\mathcal{F}_t \text{ by progr. meas.}} \right\} \in \mathcal{F}_t \text{ because:}$$

For sets of the form $[0, s] \times F$, $s \in [0, t]$, $F \in \mathcal{F}_t$, we have

$$\{\omega \in \Omega : (\tau(\omega), \omega) \in [0, s] \times F\} = \{\tau \leq s\} \cap F \in \mathcal{F}_t$$

Since $\mathcal{B}([0, t]) \times \mathcal{F}_t = \sigma([0, s] \times F, s \in [0, t], F \in \mathcal{F}_t)$, the result extends to the whole σ-algebra. ■

Proposition 13

Let $\mathbb{X}$ be continuous and $A \subseteq \mathbb{R}^d$ be an open set. Then $\tau_A := \inf\{t \geq 0 : X_t \notin A\}$ is a stopping time. If A is closed and $\bigcap_{\epsilon>0} \mathcal{F}_{t+\epsilon} = \mathcal{F}_{t^+} = \mathcal{F}_t$ then τ_A is a stopping time.

Proof

If A is open:

- by continuity of $s \mapsto X_s \implies$ continuity of $s \mapsto \text{dist}(X_s, A^c)$ and A^c is closed.

Homework 14

Fill in the details of this proof.

$$\{\tau_A > t\} = \{\inf_{0\leq s\leq t} \text{dist}(X_s, A^c) > 0\} = \bigcup_{n\in\mathbb{N}} \left\{ \inf_{0\leq s\leq t} \text{dist}(X_s, A^c) > \frac{1}{n} \right\}$$

$$\overset{\text{indistinguishability}}{=} \bigcup_{n\in\mathbb{N}} \bigcap_{s\in[0,t]\cap\mathbb{Q}} \left\{ \text{dist}(X_s, A^c) > \frac{1}{n} \right\} \in \mathcal{F}_t$$

If A is closed, let's approximate A with open sets $A_n \downarrow A$ ($A_n \supseteq A_{n+1} \supseteq A$). Then $\{\tau_A < t\} = \bigcup_{n\in\mathbb{N}}\{\tau_{A_n} \leq t\} \in \mathcal{F}_t$.

$\omega \in \{\tau_A < t\} \implies X_{\tau_A}(\omega) \in A^c \implies X_{\tau_A}(\omega) \in A_n^c$ for large n (because A^c is open and $\mathbb{X}$ continuous) $\implies \tau_{A_n}(\omega) \leq t$ for large n.

$\omega \in \bigcup_{n\in\mathbb{N}}\{\tau_{A_n} \leq t\} \implies \exists n \in \mathbb{N}$ s.t. $\omega \in \{\tau_{A_n} \leq t\} \subseteq \{\tau_A < t\}$.

Now: $\{\tau_A \leq t\} = \bigcap_{n\in\mathbb{N}} \underbrace{\{\tau_A < t + \frac{1}{n}\}}_{\in \mathcal{F}_{t+\frac{1}{n}}} \in \mathcal{F}_{t^+} = \mathcal{F}_t$. ■

2 Basic facts about Markov processes

Intermission 15

In this section we formalize the **Markov Property** in continuous time. In simple words, this property states that the future of a process depends only on its present state, not its past history.

We start by introducing the **Markov Transition Function** which governs how a process moves between states. Via the **Chapman-Kolmogorov equation** we ensure that moving from s to t is the same as stopping at $u \in (s,t)$ and restarting. We then extend this notion via the **Strong Markov Property** which says that the "memoryless" property holds even if the time we stop to check the process is random, as long as it's a stopping time.

This set-up is then used to introduce some very useful results and concepts for our modelling endeavors:

1. The solutions of SDEs are Markov processes. Hence, we can use the current state of a system to forecast future distributions without needing the entire historical data-set.
2. By transitioning from the Markov process to the **Infinitesimal Generator** and the **Semigroup** we can turn a stochastic path problem into a deterministic PDE, which we can solve via finite-difference methods. This saves us computational time, and resources, compared to using Monte Carlo methods.
3. **Càdlàg** functions are essential for modelling "shocks" or jumps, such as sudden market crashes.

Taken from Chapter 6 of Baldi 2017.

Definition 16 (Markov transition function)
Let $(E, \mathcal{E})$ be a measurable space. A MTF is a mapping $p : [0, \infty) \times [0, \infty) \times E \times \mathcal{E} \to [0, 1]$ s.t.

1. $x \mapsto p(s, t, x, A)$ is Borel measurable.
2. $A \mapsto p(s, t, x, A)$ is a probability measure on $(E, \mathcal{E})$.
3. **Chapman-Kolmogorov**: for $s < u < t$,

$$p(s, t, x, A) = \int_E p(s, u, x, dy)p(u, t, y, A).$$

Definition 17 (Markov process)
Let μ be a probability measure on $\mathcal{E}$.

A Markov process with MTF p is a stochastic process $\mathbb{X} := (\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, (X_t)_{t\geq 0}, \mathbb{P})$ s.t.

1. $\mathbb{P}(X_0 \in dy) = \mu(dy)$.

2. For every bounded measurable $f : E \to \mathbb{R}$ it holds:
$$\mathbb{E}[f(X_t)|\mathcal{F}_s] = \int_E f(y)p(s,t,X_s,dy)$$
(notice that for $f(y) = 1_A(y), A \in \mathcal{E} \implies \mathbb{P}(X_t \in A|\mathcal{F}_s) = p(s,t,X_s,A)$).

Remark 18. *In general $(\mathcal{F}_t)_{t\geq 0}$ is a filtration that contains $(\sigma(X_s, s \leq t))_{t\geq 0}$ but w.l.o.g. we could just consider $\mathcal{F}_t = \sigma(X_s, s \leq t)$.*

Homework 19
Show that $x \mapsto \int_E f(y)p(s,t,x,dy)$ is measurable.

Hint: take simple functions $f(\cdot)$. Approximate with the **Monotone Convergence Theorem**.

Theorem 20
Given a probability measure μ on $(E, \mathcal{E})$ and a MTF, there exists a Markov process with initial distribution μ and transition function p.

We want to be able to start our process from any time $s \in [0, \infty)$ and any position $x \in E$. Then we consider a measurable space $(\Omega, \mathcal{F})$ endowed with a family of filtrations $(\mathcal{F}_t^s)_{s\leq t, t\geq 0}$ and a family of probability measures $(\mathbb{P}_{s,x})_{s\geq 0, x\in E}$. Then a realisation of the Markov process reads:
$$\mathbb{X} := (\Omega, \mathcal{F}, (\mathcal{F}_t^s)_{t\geq s, s\geq 0}, (X)_{t\geq s}, (\mathbb{P}_{s,x})_{x\in E, s\geq 0})$$
and it holds:

1. $\mathbb{P}_{s,x}(X_s = x) = 1$.

2. $\mathbb{E}_{s,x}[f(X_t)|\mathcal{F}_u^s] = \int_E f(y)p(u,t,X_u,dy), \quad s \leq u \leq t$.

(notice that $\mathbb{E}_{s,x}[f(X_t)|\mathcal{F}_s^s] = \int_E f(y)p(s,t,x,dy)$). *Note: here we lose track of $s \geq 0$ then $\int_E f(y)p(u,t,X_u,dy) = \mathbb{E}_{u,z}[f(X_t)|\mathcal{F}_u^s]$ for $s < u < t$.*

Markov process on canonical space

It's convenient to define MP on a "space of paths". The regularity of paths depends on the specific applications. Generally one considers either:

(a) Continuous paths: $\Omega = C(\mathbb{R}_+; \mathbb{R}^d) \hookrightarrow \{f : \mathbb{R}_+ \to \mathbb{R}^d \text{ s.t. } f \text{ is continuous}\}$.

(b) Right-continuous with left limits (RCLL/càdlàg): $\Omega = D(\mathbb{R}_+;\mathbb{R}^d) \hookrightarrow \{f : \mathbb{R}_+ \to \mathbb{R}^d \text{ s.t. } f(t) = f(t+) \forall t \geq 0 \text{ and } f(t-) \text{ exists } \forall t > 0\}$.

- On either such space, the canonical process is

$$\begin{aligned} X : \mathbb{R}_+ \times \Omega &\to \mathbb{R}^d \\ (t,\omega) &\mapsto \omega(t) \end{aligned}$$

  here $\omega \in \Omega$ is a function $t \mapsto \omega(t)$ and $X_t(\omega) = \omega(t)$.

- We introduce a shift-operator $(\theta_h)_{h\geq 0}$ that acts on paths $\theta_h : \Omega \to \Omega$ as $\theta_h\omega(\cdot) = \omega(h+\cdot)$.

  Notice that $\theta_h X_t(\omega) = X_t(\theta_h\omega) = \omega(h+t)$.

- The filtration is the canonical filtration $\mathcal{F}_t^s = \sigma(X_u, s \leq u \leq t)$.

- MP on canonical space:

$$\mathbb{X} = (\Omega, \mathcal{F}, (\mathcal{F}_t^s)_{t\geq s, s\geq 0}, (X_t), (\theta_t), (\mathbb{P}_{s,x})_{s\geq 0, x\in\mathbb{R}^d})$$

Remark 21 (Homogeneous MP)**.** *A MTF is homogeneous if $p(s,t,x,A) = p(0,t-s,x,A)$. In this case $\mathbb{P}_{s,x}(X_t \in A) = \mathbb{P}_{0,x}(X_{t-s} \in A)$ and therefore it suffices to consider a MTF $p : \mathbb{R}_+ \times E \times \mathcal{E} \to [0,1]$, $p(t,x,A)$ and a family $(\mathbb{P}_x)_{x\in E}$.*

Something with a flavour of the freezing lemma holds:

Proposition 22
Let Y be bounded and $\sigma(X_s, s \geq t)$-measurable. Then $\mathbb{E}_{s,x}[Y|\mathcal{F}_t^s] = \mathbb{E}_{t,X_t}[Y] := (\mathbb{E}_{t,z}[Y])|_{z=X_t(\omega)}$.

Proof. The σ-algebra $\sigma(X_s, s \geq t)$ is generated by the π-system

$$\left(\{X_{t_1}^{-1}(A_1) \cap X_{t_2}^{-1}(A_2) \cap \cdots \cap X_{t_n}^{-1}(A_n)\}, \forall (t_i, A_i) \in [t,\infty) \times \mathcal{B}(\mathbb{R}^d), i = 1\ldots n, n \in \mathbb{N}\right)$$

Let $\mathcal{H}$ be the class of bounded $\sigma(X_s, s \geq t)$-measurable random variables for which it holds

$$\mathbb{E}_{s,x}[Y|\mathcal{F}_t^s] = \mathbb{E}_{t,X_t}[Y]$$

Clearly, if $(Y_n)_{n\in\mathbb{N}} \subset \mathcal{H}$, $0 \leq Y_n \leq Y_{n+1}$ and $Y_n \uparrow Y$, then $Y \in \mathcal{H}$ by MON.

Homework 23
Verify the statement above.

It's obvious that $\mathcal{H}$ is a vector space and that $Y(\omega) = 1 \in \mathcal{H}$. It remains to show that $Y(\omega) = 1_A(\omega) \in \mathcal{H}$ for any A from the π-system above.

Take $Y(\omega) = 1_{\{X_{t_1}\in A_1\}} 1_{\{X_{t_2}\in A_2\}} \dots 1_{\{X_{t_n}\in A_n\}}$.

$$\begin{aligned}
\mathbb{E}_{s,x}[Y|\mathcal{F}_t^s] &= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-1} 1_{\{X_{t_i}\in A_i\}} \cdot 1_{\{X_{t_n}\in A_n\}}\right] \\
&= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-1} 1_{\{X_{t_i}\in A_i\}} \mathbb{E}_{s,x}[1_{\{X_{t_n}\in A_n\}}|\mathcal{F}_{t_{n-1}}^s]\right] \\
&= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-1} 1_{\{X_{t_i}\in A_i\}} \underbrace{\mathbb{P}_{s,x}(X_{t_n}\in A_n|\mathcal{F}_{t_{n-1}}^s)}_{=p(t_{n-1},t_n,X_{t_{n-1}},A_n)=:\tilde{f}_{n-1}(X_{t_{n-1}})}\right] \\
&= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-2} 1_{\{X_{t_i}\in A_i\}} \underbrace{\mathbb{E}_{s,x}[1_{\{X_{t_{n-1}}\in A_{n-1}\}}\tilde{f}_{n-1}(X_{t_{n-1}})|\mathcal{F}_{t_{n-2}}^s]}_{=\int_{A_{n-1}}\tilde{f}_{n-1}(y)p(t_{n-2},t_{n-1},X_{t_{n-2}},dy)=:\tilde{f}_{n-2}(X_{t_{n-2}})}\right] \\
&= \dots \text{iterating} \dots = \mathbb{E}_{s,x}[1_{\{X_{t_1}\in A_1\}}\tilde{f}_1(X_{t_1})|\mathcal{F}_t^s] \\
&= \int_{A_1}\tilde{f}_1(y)p(t,t_1,X_t,dy) = \mathbb{E}_{t,X_t}[\tilde{f}_1(X_{t_1})1_{\{X_{t_1}\in A_1\}}] \\
&= \mathbb{E}_{t,X_t}\left[1_{\{X_{t_1}\in A_1\}}\int_E 1_{A_2}(y)\tilde{f}_2(y)p(t_1,t_2,X_{t_1},dy)\right] \\
&= \mathbb{E}_{t,X_t}\left[1_{\{X_{t_1}\in A_1\}}\mathbb{E}_{t,X_t}[1_{\{X_{t_2}\in A_2\}}\tilde{f}_2(X_{t_2})|\mathcal{F}_{t_1}^t]\right] \\
&= \mathbb{E}_{t,X_t}\left[1_{\{X_{t_1}\in A_1\}}1_{\{X_{t_2}\in A_2\}}\underbrace{\mathbb{E}_{t_2,X_{t_2}}[1_{\{X_{t_3}\in A_3\}}\tilde{f}_3(X_{t_3})|\mathcal{F}_{t_2}^t]}_{=\mathbb{E}_{t,X_t}[1_{\{X_{t_3}\in A_3\}}\tilde{f}_3(X_{t_3})|\mathcal{F}_{t_2}^t]}\right] \\
&= \dots \text{iterating} \dots = \mathbb{E}_{t,X_t}\left[\prod_{i=1}^{n} 1_{\{X_{t_i}\in A_i\}}\right] = \mathbb{E}_{t,X_t}[Y].
\end{aligned}$$

Then we can apply the MCT. □

Definition 24 (Strong Markov property)

A MP is Strong MP if for any $(\mathcal{F}_t^s)_{t\geq s}$-stopping time τ $(\tau < \infty)$

$$\mathbb{P}_{s,x}(X_{\tau+h}\in A|\mathcal{F}_\tau^s) = p(\tau,\tau+h,X_\tau,A) \quad \mathbb{P}_{s,x}\text{-a.s.}$$

Remark 25. *On the canonical space, if $Y \in m\mathcal{F}_\infty^s$ then $Y = f((X_t)_{t\geq s})$. So, given a stopping time τ we have $Y\circ\theta_\tau = f((X_t\circ\theta_\tau)_{t\geq s}) = f((X_{t+\tau})_{t\geq 0}) \in \mathcal{F}_\infty^\tau$.*

Proposition 26

If $Y \in m\mathcal{F}_\infty^s$ bounded or positive and $\mathbb{X}$ is Strong Markov, then

$$\mathbb{E}_{s,x}[Y\circ\theta_\tau|\mathcal{F}_\tau^s] = \mathbb{E}_{\tau,X_\tau}[Y] := (\mathbb{E}_{t,z}[Y])|_{t=\tau(\omega),z=X_\tau(\omega)}.$$

Proof. (HW) using the same tricks as before. □

Given a time-homogeneous MTF, the expectation is a linear operator that satisfies the semi-group property: $\forall$ bounded measurable $f : \mathbb{R}^d \to \mathbb{R}$

$$(P_t f)(x) := \int_E f(y)p(t,x,dy) = \mathbb{E}_x[f(X_t)]$$

satisfies $[P_s(P_t f)](x) = \int_E p(s,x,dy) \int_E f(z)p(t,y,dz)$

$$= \int_E \int_E f(z)p(t,y,dz)p(s,x,dy) = \mathbb{E}_x[\mathbb{E}_{X_s}[f(X_t)]]$$

$$\overset{\text{Chapman-Kolmogorov eq.}}{=} \int_E f(z)p(t+s,x,dz) = \mathbb{E}_x[f(X_{t+s})]$$

$$= (P_{t+s}f)(x)$$

Semigroup property: $P_t P_s = P_{t+s}$.

Infinitesimal generator:

$$(\mathcal{L}f)(x) := \lim_{t \downarrow 0} \frac{(P_t f)(x) - f(x)}{t} \quad \text{if the limit exists}$$

- Domain of $\mathcal{L}$, $D(\mathcal{L})$ ...
- Unbounded operator ...
- Limit in suitable norms ...

We will only be concerned with SDEs and their infinitesimal generators.

An important class of Strong MPs is the solutions of SDEs.

Stability estimates for SDEs

Here we consider $(X_t)_{t \geq 0} \in \mathbb{R}^d$ solution of

$$X_t = x + \int_0^t \mu(X_s)ds + \int_0^t \sigma(X_s)dW_s$$

for Lipschitz coefficients $\mu : \mathbb{R}^d \to \mathbb{R}^d$, $\sigma : \mathbb{R}^d \to \mathbb{R}^{d \times d'}$ and $(W_t)_{t \geq 0}$ a d'-dimensional B.m.

Standard estimates via Gronwall's inequality (HW)

(i) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_t\|^p\right] \leq C(p,T)(1 + \|x\|^p)$

(ii) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_t - x\|^p\right] \leq C(p,T)(1 + \|x\|^p) \cdot T^{p/2}$

(iii) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_t^{x_1} - X_t^{x_2}\|^p\right] \leq C(p,T)\|x_1 - x_2\|^p$

(iv) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_{t+h} - X_t\|^p\right] \leq C(p,T)(1 + \|x\|^p)h^{p/2} \quad h \geq 0$

where $p \in [2, \infty)$.

From (i) we deduce an important fact: $t \mapsto \int_0^t \sigma(X_s)dW_s$ is a martingale with

$$\mathbb{E}\left[\sup_{0 \leq t \leq T} \left| \int_0^t \sigma(X_s)dW_s \right|^p \right] < \infty$$

Moreover:

$$\mathbb{E}[\|X_t^x - X_s^y\|^p] \le C(p,T)(1+\|x\|^p+\|y\|^p)(|t-s|^{p/2}+|x-y|^p)$$

So, for $\frac{p}{2} > 1$ we can invoke Kolmogorov's continuity theorem and deduce $(x,t) \mapsto X_t^x(\omega)$ is continuous $\forall \omega \in \Omega$. (there exists a continuous modification ...) [*for time-in-homogeneous case* $(x,s,t) \mapsto X_t^{s,x}(\omega)$ *is continuous*]

Lemma 27

Let $\mathcal{H}_t := \sigma(W_u - W_t, u \ge t)$. Assume $(Y_t)_{t\ge 0} \in L^2(0,T)$ is such that $Y_u \in m\mathcal{H}_t, \forall u \ge t$. Then, $\forall v \ge t$, $\int_t^v Y_u dW_u \in \mathcal{H}_t$ and therefore $\int_t^v Y_u dW_u \perp\!\!\!\perp \sigma(W_s, s \le t)$.

Recall that Y *is* $\sigma(W_s, s\le t)$ *adapted.*

Proof. When we constructed the stoch. integral we proved that $(Y_t)_{t\ge0}$ can be approximated by simple processes $(Y_u^n)_{t\ge 0}, n\in\mathbb{N}$ s.t. $Y_u^n \in m\mathcal{H}_t, \forall u \ge t, \forall n \in \mathbb{N}$ and $\lim_{n\to\infty}\mathbb{E}[\int_0^T |Y_u^n - Y_u|^2 du] = 0$. Then, it suffices to prove the claim for simple processes $Y_u(\omega) := \sum_{i=1}^m X_{t_i}(\omega)1_{[t_i,t_{i+1})}(u)$ with $X_{t_i} \in m\mathcal{H}_t \cap m\mathcal{F}_{t_i}, \forall i = 1\dots m$ and $t \le t_1 < t_2 < \dots < t_m \le v$. For such process $\int_t^v Y_u dW_u = \sum_{i=1}^m \underbrace{X_{t_i}}_{\in m\mathcal{H}_t}\underbrace{(W_{t_{i+1}} - W_{t_i})}_{\in m\mathcal{H}_t} \in m\mathcal{H}_t$. ☐

Remark 28. *The next result holds in greater generality but we only prove it for Lipschitz coefficients.*

Proposition 29

Let $\mu : \mathbb{R}^d \to \mathbb{R}^d, \sigma: \mathbb{R}^d \to \mathbb{R}^{d\times d'}$ be Lipschitz and $(W_t)_{t\ge0} \in \mathbb{R}^{d'}$ be a B.m. Let $(X_t)_{t\ge s}$ be the unique strong solution of $X_t = x + \int_s^t \mu(X_u)du + \int_s^t \sigma(X_u)dW_u$. Denote it $(X_t^{s,x})_{t\ge s}$. Then $X_t^{s,x} \in m\mathcal{H}_s, \forall t \ge s \implies X_t^{s,x} \perp\!\!\!\perp \sigma(W_u, u \le s)$.

Proof. Recall that $(X_t^{s,x})_{t\ge s}$ is constructed by Picard iteration $X_t^0 := x, \forall t \ge 0$ and $X_t^{n+1} = x + \int_s^t \mu(X_u^n)du + \int_s^t \sigma(X_u^n)dW_u, \forall n \ge 0$. and $\lim_{n\to\infty}\mathbb{E}[\sup_{s\le t\le T}|X_t^n - X_t|^2] = 0, \forall T > 0$. $(*)$

It is clear that $X_t^0 \in m\mathcal{H}_s, \forall t \ge s$. Then $\int_s^t \mu(X_u^0)du \in m\mathcal{H}_s$ and $\int_s^t \sigma(X_u^0)dW_u \in m\mathcal{H}_s$. Now, by induction, assume $(X_t^n)_{t\ge s} \subset m\mathcal{H}_s$. Then

$$X_t^{n+1} = x + \underbrace{\int_s^t \mu(X_u^n)du}_{\in m\mathcal{H}_s} + \underbrace{\int_s^t \sigma(X_u^n)dW_u}_{\in m\mathcal{H}_s} \in m\mathcal{H}_s, \quad \forall t \ge s.$$

From $(*)$ we know $\exists (X_t^{n_j})_{t\ge s}$ s.t. $\mathbb{P}(\lim_{j\to\infty}\sup_{s\le t\le T}|X_t^{n_j} - X_t| = 0) = 1$ so $(X_t^{s,x})_{t\ge s} \in m\mathcal{H}_s$ because we assume $\mathcal{H}_s$ completed with null sets. ☐

Let's go back to considering the unique strong solution of the SDE. Since we are working with strong solutions, then $\sigma(X_r, r\le t) \subseteq \sigma(W_r, r \le t)$ and it suffices for our aims to consider $\mathcal{F}_t := \sigma(W_r, r\le t)$ (it would be equivalent to consider $\sigma(X_r, r \le t)$).

$$
\begin{aligned}
X_t^{0,x} &= x + \int_0^t \mu(X_u)du + \int_0^t \sigma(X_u)dW_u \\
&= \underbrace{x + \int_0^s \mu(X_u)du + \int_0^s \sigma(X_u)dW_u}_{=X_s^{0,x}} \\
&\quad + \int_s^t \mu(X_u)du + \int_s^t \sigma(X_u)dW_u \\
&= \underbrace{X_s^{0,x}}_{\in m\mathcal{F}_s} + \underbrace{\int_s^t \mu(X_u)du + \int_s^t \sigma(X_u)dW_u}_{\in m\mathcal{H}_s \perp\!\!\!\perp \mathcal{F}_s} =: X_s^{0,x} + Z_{s,t} = X_t^{s,X_s^{0,x}}
\end{aligned}
$$

Proposition 30

For any bounded measurable function $f : \mathbb{R}^d \to \mathbb{R}$

$$
\mathbb{E}[f(X_t)|\mathcal{F}_s] = \mathbb{E}[f(X_t)|\sigma(X_s)], \quad \forall t \geq s.
$$

Proof. From the discussion preceding the proposition we have $\mathbb{E}[f(X_t)|\mathcal{F}_s] = \mathbb{E}[f(\underbrace{X_s}_{\in m\mathcal{F}_s} + \underbrace{Z_{s,t}}_{\perp\!\!\!\perp \mathcal{F}_s})|\mathcal{F}_s]$ then we can use the freezing lemma to claim: $\mathbb{E}[f(X_s + Z_{s,t})|\mathcal{F}_s] = \psi(X_s)$ where $\psi(x) := \mathbb{E}[f(x + Z_{s,t})]$. For any $A \in \sigma(X_s)$ we have

$$
\mathbb{E}[1_A f(X_t)] = \mathbb{E}[1_A \underbrace{\mathbb{E}[f(X_t)|\mathcal{F}_s]}_{\text{by tower property}}] = \mathbb{E}[\psi(X_s)1_A]
$$

and therefore, by definition of conditional expectation,

$$
\psi(X_s) = \mathbb{E}[f(X_t)|\sigma(X_s)].
$$

Then $\mathbb{E}[f(X_t)|\sigma(X_s)] = \mathbb{E}[f(X_t)|\mathcal{F}_s]$ as claimed. □

Corollary 31

The solution $(X_t^{0,x})_{t\geq 0}$ of the SDE above is a Markov process.

Proof. Given $\Gamma \in \mathbb{R}^d$, set $f(x) = 1_\Gamma(x)$. Then

$$
\begin{aligned}
\mathbb{P}(X_t \in \Gamma|\mathcal{F}_s) &= \mathbb{P}(X_t \in \Gamma|\sigma(X_s)) = \psi(X_s) = \mathbb{P}(y + Z_{s,t} \in \Gamma)|_{y=X_s} \\
&= \mathbb{P}(X_t^{s,y} \in \Gamma)|_{y=X_s} \doteq p(s,t,y,\Gamma)|_{y=X_s} = p(s,t,X_s,\Gamma).
\end{aligned}
$$

□

Remark 32. *For any bounded and continuous $f : \mathbb{R}^d \to \mathbb{R}$ we have, by continuity of $(t,s,x) \mapsto X_s^{t,x}(\omega)$ and DOM:*

$$\begin{aligned}\int_{\mathbb{R}^d} f(y)p(t,s,x,dy) &= \mathbb{E}[f(X_s^{t,x})] \\ &= \mathbb{E}[\lim_{n\to\infty} f(X_{s_n}^{t_n,x_n})] = \lim_{n\to\infty} \mathbb{E}[f(X_{s_n}^{t_n,x_n})] \\ &= \lim_{n\to\infty} \int_{\mathbb{R}^d} f(y)p(t_n,s_n,x_n,dy) \quad \text{for any } (t_n,s_n,x_n) \to (t,s,x)\end{aligned}$$

That is, **the Feller property** *holds:*

$$(t,x,s) \mapsto \int_{\mathbb{R}^d} f(y)p(t,s,x,dy) \text{ is continuous } \forall f \in C_b(\mathbb{R}^d).$$

It thus follows that $(X_t)_{t\geq 0}$ is actually Strong-Markov!

Proposition 33

The solution $(X_t)_{t\geq 0}$ of the SDE is strong Markov.

Proof. Let τ be a (bounded) stopping time and f a bounded measurable function. We must prove that for any $\Gamma \in \mathcal{F}_\tau$ it holds (notice that $t \mapsto X_t(\omega)$ is progr. meas $\implies X_\tau(\omega) \in m\mathcal{F}_\tau$)

$$\mathbb{E}[1_\Gamma 1_{\{X_{\tau+h}\in A\}}] = \mathbb{E}[1_\Gamma p(\tau, \tau+h, X_{\tau+h}, A)].$$

We can approximate τ with a decreasing sequence of stopping times $\tau_n \downarrow \tau$ as $n \to \infty$ with τ_n taking finitely many values.

$$\begin{aligned}\mathbb{E}[1_\Gamma 1_{\{X_{\tau_n+h}\in A\}}] &= \sum_{i=1}^N \mathbb{E}[1_\Gamma 1_{\{X_{t_i+h}\in A\}} 1_{\{\tau_n = t_i\}}] \\ &= \sum_{i=1}^N \mathbb{E}[\mathbb{E}[\underbrace{1_\Gamma \cap \{\tau_n = t_i\}}_{\in \mathcal{F}_{t_i} \text{ because } \{\tau_n = t_i\} \subseteq \{\tau \leq t_i\} \text{ and } \Gamma \in \mathcal{F}_\tau} 1_{\{X_{t_i+h}\in A\}} | \mathcal{F}_{t_i}]] \\ &= \sum_{i=1}^N \mathbb{E}[1_{\Gamma\cap\{\tau_n = t_i\}} \mathbb{P}(X_{t_i+h} \in A | \mathcal{F}_{t_i})] \\ &= \sum_{i=1}^N \mathbb{E}[1_{\Gamma\cap\{\tau_n = t_i\}} p(t_i, t_i+h, X_{t_i+h}, A)] \\ &= \sum_{i=1}^N \mathbb{E}[1_{\Gamma\cap\{\tau_n = t_i\}} p(\tau_n, \tau_n+h, X_{\tau_n+h}, A)] \\ &= \mathbb{E}[1_\Gamma p(\tau_n, \tau_n+h, X_{\tau_n+h}, A)]\end{aligned}$$

Now, for any continuous, bounded function,

$$\mathbb{E}[1_\Gamma f(X_{\tau_n+h})] = \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau_n, \tau_n+h, X_{\tau_n+h}, dy)]$$

Using DOM and Feller property we have

$$\begin{aligned}
\mathbb{E}[1_\Gamma f(X_{\tau+h})] &= \lim_{n\to\infty} \mathbb{E}[1_\Gamma f(X_{\tau_n+h})] \\
&= \lim_{n\to\infty} \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau_n, \tau_n + h, X_{\tau_n+h}, dy)] \\
&= \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau, \tau + h, X_{\tau+h}, dy)]
\end{aligned}$$

$$\implies \mathbb{E}[1_\Gamma f(X_{\tau+h})] = \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau, \tau + h, X_{\tau+h}, dy)] \quad (*)$$

Any bounded measurable function can be approximated by bounded continuous functions: $f \in B_b(\mathbb{R}^d) \implies \exists (f_n)_n \subseteq C_b(\mathbb{R}^d)$ s.t. $\|f_n\|_\infty < 1 + \|f\|_\infty$ and $f_n(x) \to f(x)$ for all $x \in \mathbb{R}^d$. Then $(*)$ extends to bounded measurable functions by approximation. $\square$

3 Optimal Stopping in Discrete Time

Taken from Chapter 1 of Peskir and Shiryaev 2006.

3.1 Introduction to Optimal Stopping

- Discrete time setting: $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$
- Gain process: $(G_n)_n$ adapted to $(\mathcal{F}_n)_n$
- Finite-time horizon: N
- Class of stopping times: $\mathcal{T} := \{\tau : \tau : \Omega \to \mathbb{Z}^+ \text{ is stopping time}\}$
- For $n \leq N$, $\mathcal{T}_{n,N} := \{\tau \in \mathcal{T} : n \leq \tau \leq N\}$

The problem of interest: $V_* = \sup_\tau \mathbb{E}[G_\tau]$. We want to:

1. Calculate V_*
2. Determine optimal τ_*

We distinguish two cases:

- **Infinite-time horizon:** $\tau \in \mathcal{T}$. In this case $N = +\infty$ and we set $G_N(\omega) \equiv 0$. We need to define $G_\tau 1_{\{\tau=+\infty\}}$. The usual choice is:
$$G_\tau 1_{\{\tau=+\infty\}} := \limsup_{n\to\infty} G_n 1_{\{\tau=+\infty\}}$$
- **Finite-time horizon:** $\tau \in \mathcal{T}_{0,N}$.

Assumption:
$$\mathbb{E}\left[\sup_{0\leq n\leq N} |G_n|\right] < \infty$$

Dynamic view of the problem: $V_n^N := \sup_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[G_\tau]$.

3.2 Method of backward induction

It works for $N < \infty$.

Set $S_N^N(\omega) := G_N(\omega)$ and
$$S_n^N(\omega) := \max\left\{G_n(\omega), \mathbb{E}[S_{n+1}^N|\mathcal{F}_n](\omega)\right\}$$
for $n \in \{0, 1, \ldots, N-1\}$. Let $\tau_n^N := \inf\{n \leq k \leq N : S_k^N = G_k\}$ and notice that $S_N^N = G_N \implies \{n \leq k \leq N : S_k^N = G_k\} \neq \emptyset$.

Theorem 34 (OS1)
$\forall 0 \le n \le N$ we have, $\mathbb{P}$-a.s.

$$\left.\begin{aligned} S_n^N &\ge \mathbb{E}[G_\tau|\mathcal{F}_n] \quad \forall \tau \in \mathcal{T}_{n,N} \\ S_n^N &= \mathbb{E}[G_{\tau_n^N}|\mathcal{F}_n] \end{aligned}\right\} \quad (A) \tag{1}$$

Moreover:

(i) τ_n^N is optimal for V_n^N, i.e., $V_n^N = \mathbb{E}[G_{\tau_n^N}]$.

(ii) If $\tau_* \in \mathcal{T}_{n,N}$ is another optimal stopping time for V_n^N, then $\mathbb{P}(\tau_* \ge \tau_n^N) = 1$ (i.e., τ_n^N is **minimal**).

(iii) $(S_k^N)_{n\le k\le N}$ is the smallest supermartingale which dominates $(G_k)_{n\le k\le N}$.

(iv) $(S_{k\wedge\tau_n^N}^N)_{n\le k\le N}$ is a martingale.

Proof. **Proof of (A):** For $n = N$ we have $S_N^N = G_N$ and $\tau_N^N = N$ so the claim holds. Now assume the claim holds for generic n. Take $\tau \in \mathcal{T}_{n-1,N}$ and set $\bar\tau := \tau \vee n$. Then $\bar\tau \in \mathcal{T}_{n,N}$ and

$$\begin{aligned} \mathbb{E}[G_\tau|\mathcal{F}_{n-1}] &= \mathbb{E}[G_\tau 1_{\{\tau=n-1\}}|\mathcal{F}_{n-1}] + \mathbb{E}[G_\tau 1_{\{\tau>n-1\}}|\mathcal{F}_{n-1}] \\ &= G_{n-1}1_{\{\tau=n-1\}} + \mathbb{E}[G_{\bar\tau}|\mathcal{F}_{n-1}]1_{\{\tau>n-1\}} \end{aligned}$$

By definition $S_{n-1}^N \ge G_{n-1}$. By the **induction hypothesis**:

$$\mathbb{E}[G_{\bar\tau}|\mathcal{F}_{n-1}] = \mathbb{E}[\mathbb{E}[G_{\bar\tau}|\mathcal{F}_n]|\mathcal{F}_{n-1}] \le \mathbb{E}[S_n^N|\mathcal{F}_{n-1}]$$

because $\bar\tau \in \mathcal{T}_{n,N}$. Combining the above:

$$\begin{aligned} \mathbb{E}[G_\tau|\mathcal{F}_{n-1}] &\le S_{n-1}^N 1_{\{\tau=n-1\}} + \underbrace{\mathbb{E}[S_n^N|\mathcal{F}_{n-1}]}_{\le S_{n-1}^N \text{ by def of } (S_n^N)_n} 1_{\{\tau>n-1\}} \\ &\le S_{n-1}^N \quad \checkmark \end{aligned}$$

Now we check the equality for τ_{n-1}^N:

$$\mathbb{E}[G_{\tau_{n-1}^N}|\mathcal{F}_{n-1}] = G_{n-1}1_{\{\tau_{n-1}^N=n-1\}} + \mathbb{E}[G_{\tau_{n-1}^N}1_{\{\tau_{n-1}^N>n-1\}}|\mathcal{F}_{n-1}]$$

On $\{\tau_{n-1}^N = n-1\}$ we have $G_{n-1} = S_{n-1}^N$. On $\{\tau_{n-1}^N > n-1\}$ we have $\tau_{n-1}^N = \tau_n^N$ and $S_{n-1}^N > G_{n-1}$. The latter implies $S_{n-1}^N = \mathbb{E}[S_n^N|\mathcal{F}_{n-1}]$ by definition of $(S_n^N)_n$.

Thus:

$$
\begin{aligned}
\mathbb{E}[G_{\tau^N_{n-1}}|\mathcal{F}_{n-1}] &= S^N_{n-1}1_{\{\tau^N_{n-1}=n-1\}} + \underbrace{\mathbb{E}[G_{\tau^N_n}|\mathcal{F}_{n-1}]}_{=\mathbb{E}[\mathbb{E}[G_{\tau^N_n}|\mathcal{F}_n]|\mathcal{F}_{n-1}]} \; 1_{\{\tau^N_{n-1}>n-1\}} \\
&= S^N_{n-1}1_{\{\tau^N_{n-1}=n-1\}} + \underbrace{\mathbb{E}[S^N_n|\mathcal{F}_{n-1}]}_{\text{by inductive hypothesis}} \; 1_{\{\tau^N_{n-1}>n-1\}} \\
&= S^N_{n-1}1_{\{\tau^N_{n-1}=n-1\}} + \underbrace{\mathbb{E}[S^N_n|\mathcal{F}_{n-1}]}_{S^N_{n-1}} 1_{\{\tau^N_{n-1}>n-1\}} \\
&= S^N_{n-1} \quad \checkmark
\end{aligned}
$$

Now we prove (i). Since $S^N_n \geq \mathbb{E}[G_\tau|\mathcal{F}_n] \quad \forall \tau \in \mathcal{T}_{n,N}$, then

$$\mathbb{E}[S^N_n] \geq \mathbb{E}[G_\tau] \quad \forall \tau \in \mathcal{T}_{n,N} \implies \mathbb{E}[S^N_n] \geq V^N_n.$$

However, $S^N_n = \mathbb{E}[G_{\tau^N_n}|\mathcal{F}_n] \implies \mathbb{E}[S^N_n] = \mathbb{E}[G_{\tau^N_n}] \leq V^N_n$. Thus $\boxed{V^N_n = \mathbb{E}[G_{\tau^N_n}] = \mathbb{E}[S^N_n]}$ and τ^N_n is **optimal**.

Now we prove (iii). By definition $S^N_n \geq \mathbb{E}[S^N_{n+1}|\mathcal{F}_n]$ and therefore $(S^N_n)_n$ is a supermartingale. Obviously $S^N_n \geq G_n$. Let $(\tilde{S}_n)_n$ be another supermartingale s.t. $\tilde{S}_n \geq G_n \forall n$. Then $\tilde{S}_N \geq G_N = S^N_N$. Assume $\tilde{S}_k \geq S^N_k$ for $n+1 \leq k \leq N$. Then

$$S^N_n = \max\{G_n, \mathbb{E}[S^N_{n+1}|\mathcal{F}_n]\} \leq \max\{\underbrace{G_n}_{\leq \tilde{S}_n}, \underbrace{\mathbb{E}[\tilde{S}_{n+1}|\mathcal{F}_n]}_{\leq \tilde{S}_n}\} \leq \tilde{S}_n. \quad \checkmark$$

Now we prove (ii). Let $\tau_* \in \mathcal{T}_{n,N}$ be optimal. Then $V^N_n = \mathbb{E}[G_{\tau_*}]$. However $S^N_k \geq G_k, n \leq k \leq N$ implies $S^N_{\tau_*} \geq G_{\tau_*}$. Arguing by contradiction, assume $\mathbb{P}(S^N_{\tau_*} > G_{\tau_*}) > 0$. Then $\mathbb{E}[S^N_{\tau_*} - G_{\tau_*}] > 0$ and

$$V^N_n = \mathbb{E}[G_{\tau_*}] < \mathbb{E}[S^N_{\tau_*}] = \mathbb{E}[\mathbb{E}[S^N_{\tau_*}|\mathcal{F}_n]] \leq \mathbb{E}[S^N_n] = V^N_n$$

where the inequality follows by (iii) and optional sampling (τ_* is bounded). Thus $V^N_n < V^N_n$, which is impossible. So, it must be $S^N_{\tau_*} = G_{\tau_*}$, $\mathbb{P}$-a.s. This implies $\tau_* \geq \tau^N_n$ because $\tau_* \in \mathcal{T}_{n,N}$.

Now we prove (iv). Take $n \leq k \leq N-1$.

$$
\begin{aligned}
\mathbb{E}[S^N_{(k+1)\wedge\tau^N_n}|\mathcal{F}_k] &= \mathbb{E}[S^N_{\tau^N_n}1_{\{\tau^N_n \leq k\}} + S^N_{k+1}1_{\{\tau^N_n \geq k+1\}}|\mathcal{F}_k] \\
&= \mathbb{E}[S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \leq k\}}|\mathcal{F}_k] + \mathbb{E}[S^N_{k+1}|\mathcal{F}_k]1_{\{\tau^N_n \geq k+1\}}
\end{aligned}
$$

where $\{\tau^N_n \geq k+1\} = \bigcap_{j=n}^k \{S^N_j > G_j\} \in \mathcal{F}_k$. On the event $\{\tau^N_n \geq k+1\}$ we have $S^N_j > G_j$ for $n \leq j \leq k$ and therefore $S^N_k = \mathbb{E}[S^N_{k+1}|\mathcal{F}_k]$ by definition. Then:

$$
\begin{aligned}
\mathbb{E}[S^N_{(k+1)\wedge\tau^N_n}|\mathcal{F}_k] &= S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \leq k\}} + S^N_k 1_{\{\tau^N_n \geq k+1\}} \\
&= S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \leq k\}} + S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \geq k+1\}} \\
&= S^N_{k\wedge\tau^N_n}. \quad \checkmark
\end{aligned}
$$

□

Remark 35. *Eq. (A) in Thm OS1 yields:*

$$S^N_n = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{n,N}} \mathbb{E}[G_\tau|\mathcal{F}_n]$$

and it characterises the so-called **Snell-envelope** *of* $(G_n)_n$*: the smallest supermartingale that dominates* $(G_n)_n$*. Notice that* $\mathcal{T}_{n,N}$ *is an uncountable set.*

3.3 Infinite Horizon

Infinite horizon: when $N = +\infty$ the backward induction fails.

Let $\mathcal{T}_n := \{\tau \in \mathcal{T} : \tau \geq n\}$ and $V_n := \sup_{\tau \in \mathcal{T}_n} \mathbb{E}[G_\tau]$.

To solve the problem we consider a process $(S_n)_{n \in \mathbb{Z}^+}$ defined as

$$S_n := \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_n} \mathbb{E}[G_\tau | \mathcal{F}_n]$$

and the stopping time

$$\tau_n := \inf\{k \geq n : S_k = G_k\} \quad \text{with } \inf \emptyset = +\infty$$

> **Homework 36**
> Prove that $\tau_n \in \mathcal{T}_n$.

We always assume $\mathbb{E}\left[\sup_{n \geq 0} |G_n|\right] < \infty$.

Theorem 37 (OS2)

The process $(S_n)_{n \in \mathbb{Z}^+}$ satisfies

$$S_n = \max\left(G_n, \mathbb{E}[S_{n+1} | \mathcal{F}_n]\right) \quad \forall n \in \mathbb{Z}^+. \tag{A}$$

Furthermore

$$\left.\begin{aligned} &S_n \geq \mathbb{E}[G_\tau | \mathcal{F}_n] \quad \forall \tau \in \mathcal{T}_n \\ &S_n = \mathbb{E}[G_{\tau_n} | \mathcal{F}_n] \quad \text{if } \mathbb{P}(\tau_n < \infty) = 1. \end{aligned}\right\} \ (B) \tag{2}$$

Finally:

(i) τ_n is optimal if $\mathbb{P}(\tau_n < \infty) = 1$.

(ii) If τ_* is an optimal stopping time, then $\mathbb{P}(\tau_* \geq \tau_n) = 1$.

(iii) $(S_n)_{n \in \mathbb{Z}^+}$ is the smallest supermartingale that dominates $(G_n)_{n \in \mathbb{Z}^+}$.

(iv) $(S_{k \wedge \tau_n})_{k \geq n}$ is a martingale.

If $\mathbb{P}(\tau_n = +\infty) > 0$ then there is no optimal stopping time which is finite with probab. 1.

Proof. Let's start from (A). Take $\tau \in \mathcal{T}_n$ and set $\bar{\tau} := \tau \vee (n+1)$. Since $\{\tau \geq n+1\} \in \mathcal{F}_n$ we have:

$$\begin{aligned}
\mathbb{E}[G_\tau|\mathcal{F}_n] &= \mathbb{E}[G_\tau 1_{\{\tau=n\}}|\mathcal{F}_n] + \mathbb{E}[G_\tau 1_{\{\tau\geq n+1\}}|\mathcal{F}_n] \\
&= G_n 1_{\{\tau=n\}} + \mathbb{E}[G_{\bar{\tau}} 1_{\{\tau\geq n+1\}}|\mathcal{F}_n] \\
&= G_n 1_{\{\tau=n\}} + 1_{\{\tau\geq n+1\}}\mathbb{E}[\quad \underbrace{\mathbb{E}[G_{\bar{\tau}}|\mathcal{F}_{n+1}]}_{\leq \operatorname{ess\,sup}_{\tau\in\mathcal{T}_{n+1}} \mathbb{E}[G_\tau|\mathcal{F}_{n+1}]} \quad |\mathcal{F}_n] \\
&\leq G_n 1_{\{\tau=n\}} + \mathbb{E}[S_{n+1}|\mathcal{F}_n]1_{\{\tau\geq n+1\}} \\
&\leq \max\{G_n, \mathbb{E}[S_{n+1}|\mathcal{F}_n]\}.
\end{aligned}$$

Since $\tau \in \mathcal{T}_n$ was arbitrary, $\operatorname{ess\,sup}_{\tau\in\mathcal{T}_n} \mathbb{E}[G_\tau|\mathcal{F}_n] \leq \max(G_n, \mathbb{E}[S_{n+1}|\mathcal{F}_n])$.

Now we prove the reverse inequality. By definition $S_n \geq G_n$ and it remains to prove $S_n \geq \mathbb{E}[S_{n+1}|\mathcal{F}_n]$. Consider the family $\{\mathbb{E}[G_\tau|\mathcal{F}_{n+1}], \tau \in \mathcal{T}_{n+1}\} =: \Gamma_{n+1}$. Given $\tau_1, \tau_2 \in \mathcal{T}_{n+1}$ we set $\tau_3 := \tau_1 1_A + \tau_2 1_{A^c}$ with

$$A := \{\omega \in \Omega : \mathbb{E}[G_{\tau_1}|\mathcal{F}_{n+1}](\omega) > \mathbb{E}[G_{\tau_2}|\mathcal{F}_{n+1}](\omega)\}.$$

Since $A \in \mathcal{F}_{n+1}$ (HW) then $\tau_3 \in \mathcal{T}_{n+1}$ (HW). Moreover

$$\mathbb{E}[G_{\tau_3}|\mathcal{F}_{n+1}] \geq \max\{\mathbb{E}[G_{\tau_1}|\mathcal{F}_{n+1}], \mathbb{E}[G_{\tau_2}|\mathcal{F}_{n+1}]\} \quad \text{(HW)}.$$

That implies: the family Γ_{n+1} is upward directed and

$$\operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{n+1}} \mathbb{E}[G_\tau|\mathcal{F}_{n+1}] =\uparrow \lim_{k\to\infty} \mathbb{E}[G_{\tau_k}|\mathcal{F}_{n+1}] \quad \text{for some } (\tau_k)_{k\in\mathbb{N}} \in \mathcal{T}_{n+1}.$$

By <u>MON</u> we get

$$\begin{aligned}
\mathbb{E}[S_{n+1}|\mathcal{F}_n] &= \mathbb{E}[\uparrow \lim_{k\to\infty} \mathbb{E}[G_{\tau_k}|\mathcal{F}_{n+1}]|\mathcal{F}_n] \\
&=\uparrow \lim_{k\to\infty} \mathbb{E}[\mathbb{E}[G_{\tau_k}|\mathcal{F}_{n+1}]|\mathcal{F}_n] \\
&=\uparrow \lim_{k\to\infty} \underbrace{\mathbb{E}[G_{\tau_k}|\mathcal{F}_n]}_{\leq S_n \text{ because } \mathcal{T}_{n+1}\subset\mathcal{T}_n} \leq S_n.
\end{aligned}$$

> **Homework 38**
> Why can we use MON?

Then $S_n \geq \max\{G_n, \mathbb{E}[S_{n+1}|\mathcal{F}_n]\}$.

Now we prove (B). The inequality holds by definition of S_n. In order to prove $S_n = \mathbb{E}[G_{\tau_n}|\mathcal{F}_n]$ we first want to prove (iv). We know that $S_n \geq \mathbb{E}[S_{n+1}|\mathcal{F}_n]$ and therefore $(S_n)_{n\in\mathbb{N}}$ is a supermartingale. So we conclude $(S_{k\wedge\tau_n})_{k\geq n}$ is a supermartingale. It remains to show $\mathbb{E}[S_{k\wedge\tau_n}] = \mathbb{E}[S_{(k+1)\wedge\tau_n}]$.

> **Homework 39**
> If $(X_n)_{n\in\mathbb{Z}^+}$ is a supermg s.t. $\mathbb{E}[X_n] = \mathbb{E}[X_{n+1}] \forall n \in \mathbb{Z}^+$ then $(X_n)_{n\in\mathbb{Z}^+}$ is a martingale.
>
> *Sol:* $X_n \geq \mathbb{E}[X_{n+1}|\mathcal{F}_n]$ so $\mathbb{E}[X_n - X_{n+1}] = \mathbb{E}[X_n - \mathbb{E}[X_{n+1}|\mathcal{F}_n]] \geq 0$ but the equality implies $X_n = \mathbb{E}[X_{n+1}|\mathcal{F}_n]$.

$$
\begin{aligned}
\mathbb{E}[S_{(k+1)\wedge\tau_n}] &= \mathbb{E}[1_{\{\tau_n\le k\}}S_{\tau_n} + 1_{\{\tau_n\ge k+1\}}S_{k+1}] \\
&= \mathbb{E}[1_{\{\tau_n\le k\}}S_{\tau_n\wedge k} + 1_{\{\tau_n\ge k+1\}}\underbrace{\mathbb{E}[S_{k+1}|\mathcal{F}_k]}_{S_k}] \\
&\qquad \text{because } \{\tau_n \ge k+1\} = \bigcap_{j=n}^{k}\{S_j > G_j\} \in \mathcal{F}_k \\
&\qquad \text{and } S_k = \mathbb{E}[S_{k+1}|\mathcal{F}_k] \text{ on that event.} \\
&= \mathbb{E}[1_{\{\tau_n\le k\}}S_{\tau_n\wedge k} + 1_{\{\tau_n\ge k+1\}}S_{k\wedge\tau_n}] = \mathbb{E}[S_{\tau_n\wedge k}].
\end{aligned}
$$

This proves (iv).

Set $G^* := \sup_{k\in\mathbb{Z}^+}|G_k|$. Then

$$
|S_n| = \left|\operatorname*{ess\,sup}_{\tau\in\mathcal{T}_n}\mathbb{E}[G_\tau|\mathcal{F}_n]\right| \le \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_n}\mathbb{E}[|G_\tau||\mathcal{F}_n] \le \mathbb{E}[G^*|\mathcal{F}_n]
$$

Therefore $(S_n)_{n\in\mathbb{Z}^+}$ is bounded by the u.i. martingale $\{\mathbb{E}[G^*|\mathcal{F}_n], n\in\mathbb{Z}^+\}$. That implies $(S_n)_{n\in\mathbb{Z}^+}$ is a u.i. supermartingale. It follows that $(S_{k\wedge\tau_n})_{k\ge n}$ is a u.i. martingale, with $\lim_{k\to\infty}S_{k\wedge\tau_n} = S_{\tau_n} = G_{\tau_n}$ on $\{\tau_n<\infty\}$. Then by the proposition on u.i. martingales ($M_n = \mathbb{E}[M_\infty|\mathcal{F}_n]$)

$$
S_n = \mathbb{E}[S_{k\wedge\tau_n}|\mathcal{F}_n] = \mathbb{E}[S_{\tau_n}|\mathcal{F}_n] = \mathbb{E}[G_{\tau_n}|\mathcal{F}_n].
$$

This proves (B).

Let's prove (i).

$$
V_n = \sup_{\tau\in\mathcal{T}_n}\mathbb{E}[G_\tau] = \sup_{\tau\in\mathcal{T}_n}\mathbb{E}[\underbrace{\mathbb{E}[G_\tau|\mathcal{F}_n]}_{\le S_n}] \le \mathbb{E}[S_n]
$$

On the other hand $V_n \ge \mathbb{E}[G_{\tau_n}] = \mathbb{E}[\mathbb{E}[G_{\tau_n}|\mathcal{F}_n]] = \mathbb{E}[S_n]$ if $\mathbb{P}(\tau_n<\infty)=1$.

Let's prove (ii). Assume τ_* is optimal, i.e. $V_n = \mathbb{E}[G_{\tau_*}]$. Since $G_k \le S_k \forall k\in\mathbb{Z}^+$ then

$$
V_n = \mathbb{E}[G_{\tau_*}] \le \mathbb{E}[S_{\tau_*}] = \mathbb{E}[\mathbb{E}[S_{\tau_*}|\mathcal{F}_n]]
$$

<u>If</u>: $\mathbb{E}[S_{\tau_*}|\mathcal{F}_n] \le S_n$ then: $(*)$

$$
V_n = \mathbb{E}[G_{\tau_*}] \le \mathbb{E}[S_{\tau_*}] \le \mathbb{E}[S_n] = V_n
$$

Therefore $\mathbb{E}[G_{\tau_*} - S_{\tau_*}] = 0 \implies S_{\tau_*} = G_{\tau_*} \implies \tau_* \ge \tau_n$.

It remains to prove (*).

$$
\begin{aligned}
\mathbb{E}[S_{\tau_*}|\mathcal{F}_n] &= \mathbb{E}[S_{\tau_*} + \mathbb{E}[G^*|\mathcal{F}_{\tau_*}] - \mathbb{E}[G^*|\mathcal{F}_{\tau_*}]|\mathcal{F}_n] \\
&= \mathbb{E}[S_{\tau_*} + G^*|\mathcal{F}_n] - \mathbb{E}[G^*|\mathcal{F}_n] \\
&= \mathbb{E}[\liminf_{k\to\infty}(\underbrace{S_{\tau_*\wedge k} + G^*}_{\ge 0})|\mathcal{F}_n] - \mathbb{E}[G^*|\mathcal{F}_n] \\
&\le \liminf_{k\to\infty}\mathbb{E}[S_{\tau_*\wedge k} + G^*|\mathcal{F}_n] - \mathbb{E}[G^*|\mathcal{F}_n] \quad \text{(Fatou's lemma)} \\
&= \liminf_{k\to\infty}\mathbb{E}[S_{\tau_*\wedge k}|\mathcal{F}_n] \le S_n, \text{ as needed.}
\end{aligned}
$$

Let's prove (iii). Assume that $(\tilde{S}_k)_{k\geq n}$ is another supermg that dominates $(G_k)_{k\geq n}$. Then

$$S_k = \mathbb{E}[G_{\tau_k}|\mathcal{F}_k] \leq \mathbb{E}[\tilde{S}_{\tau_k}|\mathcal{F}_k] \leq \tilde{S}_k.$$

Homework 40

Hint: $\tilde{S}_k \geq G_k \geq -|G_k| \geq -G^* \implies \tilde{S}_k + G^* \geq 0 \quad \forall k \geq n.$

□

Remark 41. *It is clear that* $V_n^N \leq V_n^{N+1} \leq V_n$ *and* $S_n^N \leq S_n^{N+1} \leq S_n \quad \forall N \in \mathbb{N}, n \leq N$. *(HW)*

Then $V_n^\infty := \lim_{N\to\infty} V_n^N \leq V_n$ *and* $S_n^\infty := \lim_{N\to\infty} S_n^N \leq S_n$.

If $\mathbb{P}(\tau_n < \infty) = 1$, *then*

$$\begin{aligned}
S_n = \mathbb{E}[G_{\tau_n}|\mathcal{F}_n] &= \mathbb{E}\left[G_{\tau_n} + \sup_{k\geq n}|G_k|\,\middle|\,\mathcal{F}_n\right] - \mathbb{E}\left[\sup_{k\geq n}|G_k|\,\middle|\,\mathcal{F}_n\right] \\
&= \mathbb{E}\left[\liminf_{N\to\infty}(G_{\tau_n\wedge N} + \underbrace{\sup_{k\geq n}|G_k|)}_{\geq 0}\,\middle|\,\mathcal{F}_n\right] - \mathbb{E}\left[\sup_{k\geq n}|G_k|\,\middle|\,\mathcal{F}_n\right] \\
&\overset{\text{Fatou's}}{\leq} \liminf_{N\to\infty}\mathbb{E}\left[G_{\tau_n\wedge N} + \sup_{k\geq n}|G_k|\,\middle|\,\mathcal{F}_n\right] - \mathbb{E}\left[\sup_{k\geq n}|G_k|\,\middle|\,\mathcal{F}_n\right] \\
&= \liminf_{N\to\infty}\mathbb{E}[G_{\tau_n\wedge N}|\mathcal{F}_n] \leq \liminf_{N\to\infty} S_n^N, \quad \textit{because } \tau_n\wedge N \in \mathcal{T}_{n,N}
\end{aligned}$$

Then $S_n = \lim_{N\to\infty} S_n^N$ *and by analogous arguments* $V_n = \lim_{N\to\infty} V_n^N$.

Notice also that $\tau_n^N = \inf\{k \geq n : S_k^N = G_k\} \leq \tau_n^{N+1} \leq \tau_n \quad \forall N \in \mathbb{N}$ *because* $Y_k^N := S_k^N - G_k \geq 0$ *and* $Y_k^N \leq Y_k^{N+1}$.

Therefore $\tau_n^\infty := \lim_{N\to\infty}\tau_n^N$ *is well-defined and* $\tau_n^\infty \leq \tau_n$.

Assume $\mathbb{P}(\tau_n - \tau_n^\infty > 0) > 0$. *Then* $\mathbb{P}(\tau_n - \tau_n^\infty \geq 1) > \delta$ *for some* $\delta > 0$. *Let* $A := \{\tau_n - \tau_n^\infty \geq 1\}$. *Then for* $\omega \in A$ *we must have* $\tau_n^\infty(\omega) \leq T_\omega$ *for some* $T_\omega \in \mathbb{N}$.

$S_k(\omega) - G_k(\omega) > 0, \forall n \leq k \leq \tau_n^\infty(\omega)$. *That implies*

$$S_k(\omega) - G_k(\omega) \geq \eta_\omega > 0, \forall n \leq k \leq \tau_n^\infty(\omega), \ \textit{for some } \eta_\omega > 0.$$

In other words

$$\min_{n\leq k\leq\tau_n^\infty(\omega)}(S_k(\omega) - G_k(\omega)) \geq \eta_\omega \tag{1}$$

Since $S_k^N(\omega) \nearrow S_k(\omega)$ *as* $N \nearrow \infty \quad \forall n \leq k \leq T_\omega$ *then*

$$\max_{n\leq k\leq T_\omega}(S_k(\omega) - S_k^N(\omega)) \leq \eta_\omega/2 \quad \textit{for sufficiently large } N.$$

(Note: T_ω *is finite! i.e.,* $\exists \bar{N}_{\omega,\eta,T} \in \mathbb{N}$ *s.t.* $\max_{n\leq k\leq T_\omega}(S_k(\omega) - S_k^N(\omega)) \leq \eta_\omega/2 \quad \forall N \geq \bar{N}_{\omega,\eta,T}$*).*

Therefore, from (1) *we deduce,* $\forall N \geq \bar{N}_{\omega,\eta,T}$

$$\min_{n\leq k\leq \tau_n^\infty(\omega)} (S_k^N(\omega) - G_k(\omega)) \geq \eta_\omega/2 \quad (HW)$$

which would imply the absurd $\tau_n^N(\omega) \geq \tau_n^\infty(\omega) + 1$*. Thus* $\mathbb{P}(\tau_n - \tau_n^\infty \geq 1) = 0$ *and* $\tau_n = \tau_n^\infty$ $\mathbb{P}$*-a.s.*

3.4 Markovian Setting

Let $(X_n)_{n\in\mathbb{Z}^+}$ be a Markov process taking values on some metric space $(E, \mathrm{dist}(\cdot))$ [For example $(\mathbb{R}^m, \|\cdot\|_{\mathbb{R}^m})$].

This allows us to use the Markovian setup:

$$(\Omega, \mathcal{F}, (\mathcal{F}_n^X)_{n\in\mathbb{Z}^+}, (X_n)_{n\in\mathbb{Z}^+}, (\theta_n)_{n\in\mathbb{Z}^+}, (\mathbb{P}_x)_{x\in E})$$

where θ_n is the shift operator.

Assume that $G_n = g(X_n)$ for some measurable $g : E \to \mathbb{R}$.

3.5 Finite horizon

Then $S_N^N = g(X_N)$. Arguing by induction, assume $S_{n+1}^N = u(n+1, X_{n+1})$ for some function $u(\cdot,\cdot)$. Then

$$S_n^N = \max\left\{ g(X_n), \quad \underbrace{\mathbb{E}[u(n+1, X_{n+1})|\mathcal{F}_n]}_{f(n,X_n) \text{ by Markovianity for some } f(\cdot)} \right\}$$

$\implies S_n^N = u(n, X_n)$ with

$$u(n,x) := \max\{g(x), (P_{n+1}u(n+1,\cdot))(x)\}$$

Here we use the semigroup $(P_n f)(x) := \int f(y)p_n(x, dy)$ where $p_n(x, dy) = \mathrm{Law}(X_n|X_{n-1} = x)$.

Recall that $S_n^N = \operatorname{ess\,sup}_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[G_\tau|\mathcal{F}_n]$ and therefore

$$u(n, X_n) = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[g(X_\tau)|\mathcal{F}_n]$$

Moreover we know that $\tau_n^* = \inf\{k \geq n : u(k, X_k) = g(X_k)\}$ is optimal (HW). It is not hard to see that τ_n^* is a stopping time for the filtration generated by X, i.e. $\mathcal{F}_k^X = \sigma(X_0, X_1, \ldots, X_k)$. Then, with no loss of generality we can assume $(\mathcal{F}_k)_{k\in\mathbb{Z}^+} = (\mathcal{F}_k^X)_{k\in\mathbb{Z}^+}$.

In order to pass to the infinite-time horizon we need to change our notation slightly:

$$\boxed{S_n^N = u^N(n, X_n) = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[g(X_\tau)|\mathcal{F}_n]}$$

3.6 Infinite horizon

Let us now assume that $(X_n)_{n\in\mathbb{Z}^+}$ is a homogeneous Markov chain with $X_0 = x \in E$, so that $P_n = P$ and $(Pf)(x) = \int f(y)p(x, dy)$.

It follows that, for $X_0 = x \in E$,

$$u^N(n, X_n) = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{n,N}} \underbrace{\mathbb{E}_x[g(X_{n+\tau\circ\theta_n})|\mathcal{F}_n]}_{\overset{\text{using the shift operator}}{=} \mathbb{E}_y[g(X_\tau)]\big|_{y=X_n(\omega)}}$$

$$\text{where } \tau(\omega) = n + \tau(\theta_n(\omega)) = n + \tau \circ \theta_n$$

$$= \operatorname*{ess\,sup}_{\sigma \in \mathcal{T}_{0,N-n}} \mathbb{E}_{X_n}[g(X_\tau)] = u^{N-n}(0, X_n).$$

Then

$$\begin{aligned} u^N(0,x) &= \max\{g(x), (Pu^N(1,\cdot))(x)\} \\ &= \max\{g(x), (Pu^{N-1}(0,\cdot))(x)\}. \end{aligned} \tag{1}$$

It is clear that $u^N \leq u^{N+1}$ (HW) and therefore the limit

$$u^\infty(x) := \lim_{N\to\infty} u^N(0,x) \quad \text{exists (pointwise).}$$

Taking limits in eq. (1) we get

$$u^\infty(x) = \max\left\{g(x), \lim_{N\to\infty} \mathbb{E}_x[u^{N-1}(0, X_1)]\right\}$$

Notice that

$$\begin{aligned} u^{N-1}(0,x) &= \sup_{\tau\in\mathcal{T}_{0,N-1}} \mathbb{E}_x[g(X_\tau)] \\ &= \sup_{\tau\in\mathcal{T}_{0,N-1}} \mathbb{E}_x[g(X_\tau) + \sup_{n\in\mathbb{Z}^+} |g(X_n)| - \sup_{n\in\mathbb{Z}^+} |g(X_n)|] \\ &= \sup_{\tau\in\mathcal{T}_{0,N-1}} \mathbb{E}_x\underbrace{[g(X_\tau) + \sup_{n\in\mathbb{Z}^+} |g(X_n)|]}_{\geq 0} - \underbrace{\mathbb{E}_x[\sup_{n\in\mathbb{Z}^+} |g(X_n)|]}_{=:c(x)>0} \end{aligned}$$

Then $u^{N-1}(0,x) \geq -c(x)$ (i.e., it is bounded from below). We can apply MON to obtain

$$\boxed{u^\infty(x) = \max\{g(x), \mathbb{E}_x[u^\infty(X_1)]\}.}$$

Thanks to Thm. OS2 we can identify

$$\boxed{u^\infty(x) = V_0 = \sup_{\tau\in\mathcal{T}_0} \mathbb{E}_x[g(X_\tau)] =: V(x)}$$

Moreover, by the Markovian structure

$$V(X_n) = \left(\sup_{\tau\in\mathcal{T}_0} \mathbb{E}_y[g(X_\tau)]\right)\Big|_{y=X_n} = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_n} \mathbb{E}_x[g(X_\tau)|\mathcal{F}_n]$$

Remark 42. *This is a bit imprecise but essentially true.*

4 Stochastic Control in Discrete Time

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